# Exact Organization of Density Matrices and Entanglement Structure in the Kitaev Spin Liquid

Chen-Chih Wang, Yi-Ping Huang, Sungkit Yip, arXiv:2605.18002

Chen-Chih Wang

NTHU

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# Outline

# Motivation

- density matrix
- entanglement
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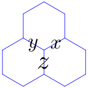
# Kitaev Model: an Introduction

- Model Hamiltonian

$$H = -\sum_{\langle ij \rangle} J_{\alpha_{ij}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \quad ;$$




- Majorana fermion (parton) representation: $b^x$, $b^y$, $b^z$, $c \Rightarrow \sigma^\mu = i b^\mu c$
- Parton Hamiltonian: define $i b_i^{\alpha_{ij}} b_j^{\alpha_{ij}} = \hat{u}_{ij}$

$$H \to \sum_{\langle ij \rangle} i J_{\alpha_{ij}} \hat{u}_{ij} c_i c_j$$

- Gutzwiller projection

$$\mathcal{P}_G = \prod_i \left( \frac{\mathbb{1} + D_i}{2} \right) \quad , \quad D_i = b_i^x b_i^y b_i^z c_i \Leftarrow -i \sigma_i^x \sigma_i^y \sigma_i^z$$

- Separability of the entanglement entropy (Yao and Qi, PRL**105** 080501)

$$S_A = S_A^F + S_A^G$$

# Kitaev Model: Gauge Theory

In the Majorana fermion representation, the model is equivalent to a system of free Majorana fermions (matter field) coupled to a static $\mathbb{Z}_2$ gauge field

- Define the gauge field $\hat{u}_{ij} = e^{iA_{ij}}$, then the elementary $\mathbb{Z}_2$ gauge flux reads

$$e^{i\Phi} = \exp\left(i \oint_{\partial p} A\right) = \prod_{\langle ij\rangle \in \partial p} \hat{u}_{ij} = \prod_{\langle ij\rangle \in \partial p} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} =: W_p$$

- The electric field in the gauge field that flips the gauge field is $b_i^{\alpha_{ij}} \to e^{i\pi E_{ij}}$ because $b_i^{\alpha_{ij}} \hat{u}_{ij} b_i^{\alpha_{ij}} = -\hat{u}_{ij}$
- Gutzwiller projection as a Gauss law constraint: $D_i = b_i^x b_i^y b_i^z c_i = 1$. The ground state is

$$\prod_i \left(\frac{\mathbb{1} + D_i}{2}\right) |u\rangle\, |M\rangle \quad ; \quad \text{Fix the gauge: } \langle u|\, \hat{u}_{ij}\, |u\rangle = +1 \forall \langle ij\rangle$$

- Both definitions $b_i^{\alpha_{ij}} = e^{i\pi E_{ij}}$ and $b_j^{\alpha_{ij}} = e^{i\pi E_{ij}}$ are valid.
- The Gauss law is a physical constraint and cannot be broken.

# Kitaev Model: Properties of Spin Correlation Functions

- Two-spin correlation functions are short-ranged (Baskaran, Mandal, and Shankar, PRL**98**, 247201)

$$\left\langle \sigma_i^\mu \sigma_j^\nu \right\rangle = \langle M| ic_i c_j |M\rangle \quad \underbrace{\langle u| ib_i^\mu b_j^\nu |u\rangle}_{\text{vanishes if } ib_i^\mu b_j^\nu \neq \hat{u}_{ij}}$$

- Multi-spin correlation functions can ONLY be string-like

$$\left\langle \prod_{\langle ij\rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right\rangle \equiv \langle \Sigma_{\mathfrak{D}} \rangle$$

- $\mathbb{Z}_2$ algebra of strings

$$\mathfrak{D} \ominus \mathfrak{D}' := \mathfrak{D} \cup \mathfrak{D}' - \mathfrak{D} \cap \mathfrak{D}' \quad : \quad \textcolor{red}{illustration}$$

- String operators form a projective representation

$$\boxed{\Sigma_{\mathfrak{D}} \Sigma_{\mathfrak{D}'} = \varphi(\mathfrak{D}, \mathfrak{D}') \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'}} \quad ; \quad \varphi(\mathfrak{D}, \mathfrak{D}') = \pm 1, \quad \varphi(\mathfrak{D}, \mathfrak{D}) = (-1)^{|\mathfrak{D}|}$$

# Organization: Equivalence Relation

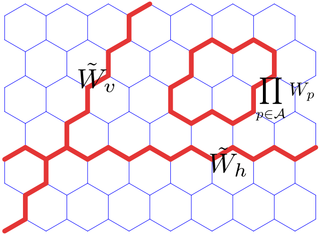
- Exact 1-form symmetry: Wilson loop $\Sigma_{\mathcal{C}} |\psi_0\rangle = W_{\mathcal{C}} |\psi_0\rangle = |\psi_0\rangle$
- The weights are identical up to a sign if the strings are identical up to a loop $\mathcal{C}$

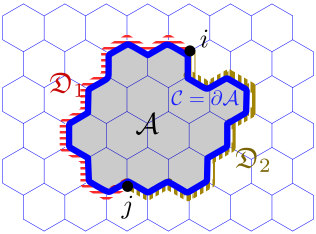
$$\langle \Sigma_{\mathfrak{D}\ominus\mathcal{C}} \rangle = \varphi(\mathfrak{D},\mathcal{C}) \langle \Sigma_{\mathfrak{D}} \Sigma_{\mathcal{C}} \rangle = \varphi(\mathfrak{D},\mathcal{C}) \langle \Sigma_{\mathfrak{D}} W_{\mathcal{C}} \rangle = \varphi(\mathfrak{D},\partial A) \langle \Sigma_{\mathfrak{D}} \rangle$$

- Equivalence relation:

$$\boxed{[\mathfrak{D}] = \left\{ \mathfrak{D}' \;:\; \exists\, \mathcal{C} \text{ composed by closed loops such that } \mathfrak{D}' \ominus \mathcal{C} = \mathfrak{D} \right\}}$$

- Endpoints perspective: $\mathrm{EP}(\mathfrak{D}) = \mathrm{EP}(\mathfrak{D}')$ iff $[\mathfrak{D}] = [\mathfrak{D}']$ $\Rightarrow$ Emerging fermions at endpoints of strings



# Organization of Density Matrix

- Suppose the observables are a set of $\mathbb{Z}_2$ traceless Hermitian operators: $\{\mathcal{O}_n^2 = \mathbb{1}\}$, then the density matrix can be expanded in terms of these observables

$$\rho \propto \sum_n \mathrm{Tr}(\mathcal{O}_n^\dagger \rho)\mathcal{O}_n = \sum_n \langle \mathcal{O}_n \rangle \mathcal{O}_n$$

- For the spin-$1/2$ Kitaev model, the observables are Pauli strings.

$$\rho \propto \sum_{\mathfrak{D}} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}}$$

- Organization due to the exact 1-form Wilson symmetry:

$$\rho \propto \mathcal{P}^{(p)} \sum_{\color{red}{[\mathfrak{D}]}} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}} \quad ; \quad \mathcal{P}^{(p)} := \underbrace{\left(\frac{\mathbb{1} + \tilde{W}_h}{2}\right)\left(\frac{\mathbb{1} + \tilde{W}_v}{2}\right)}_{\text{topological projector (global Wilson)}} \underbrace{\prod_p \left(\frac{\mathbb{1} + W_p}{2}\right)}_{\text{flux-free projector}}$$

- Normalize $\mathrm{Tr}\rho = 1$ to get the final result:

$$\boxed{\rho = \frac{1}{2^{N-N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}}}$$

# Organization of Reduced Density Matrix

- Following the same spirit to construct the reduced density matrix:
  - (i) Hermiticity
  - (ii) Ability of reproducing expectation values of observables
  - (iii) Normalization condition $\mathrm{Tr}\rho = 1$

$$\rho_A \propto \sum_{\mathfrak{D}\subset A} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}} \propto \mathcal{P}_A^{(p)} \sum_{[\mathfrak{D}]\subset A} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}} \xrightarrow{\text{normalize}} \boxed{\frac{1}{2^{N^A - N_p^A}} \mathcal{P}_A^{(p)} \sum_{[\mathfrak{D}]\subset A} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}}}$$

- Define *projected string operators*: $\mathcal{S}_{\mathfrak{D}}^{(A)} := \mathcal{P}_A^{(p)} \Sigma_{\mathfrak{D}}$. In this way, $\mathcal{S}_{\mathfrak{D}}^{(A)} = \mathcal{S}_{\mathfrak{D}'}^{(A)}$ if $\mathfrak{D} \sim \mathfrak{D}'$.
- Representing density matrices in terms of projected string operators:

$$\boxed{\rho_A = \frac{1}{2^{N^A - N_p^A}} \sum_{[\mathfrak{D}]\subset A} \langle \Sigma_{\mathfrak{D}} \rangle \mathcal{S}_{\mathfrak{D}}^{(A)}}$$

# Organization: Block-Diagonal Structure

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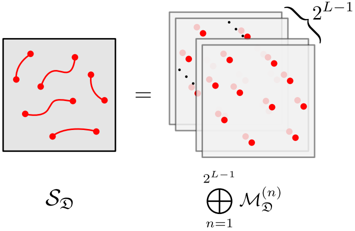


# Frame Title

- Exact Gauss law at the boundary: Block-diagonal structure
- Exact 1-form Wilson symmetry: Making blocks identical

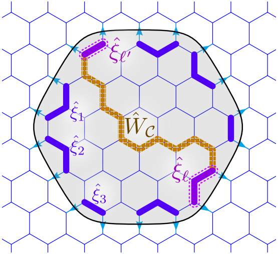
$$\Sigma_{\mathcal{C}} = \prod_{\langle ij\rangle \in \mathcal{C}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}}$$

$\sigma_1^{\mu_1}$ $\sigma_2^{\mu_2}$

$$\Updownarrow$$

$$\prod_{\langle ij\rangle \in \mathcal{C}} \hat{u}_{ij}$$

$b_1^{\mu_1}$ $b_2^{\mu_2}$



# Frame Title

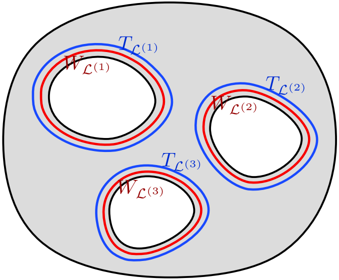
# Unsolved Problems and Outlooks

- To get the REAL topological entanglement entropy, we need an exact ’t Hooft symmetry and make it spontaneously broken.
- Higher spin Kitaev model:

# Frame Title

- Reduced density matrix of a Kitaev ground state

$$\rho_A = \frac{1}{2^{N^A - N_p^A - N_s^A - 2g}} \prod_{n=1}^{g} \left( \frac{\mathbb{1} + T_{\mathcal{L}(n)}}{2} \right) \prod_{n=1}^{g} \left( \frac{\mathbb{1} + W_{\mathcal{L}(n)}}{2} \right) \prod_{p \in A} \left( \frac{\mathbb{1} + P_p}{2} \right) \prod_{s \in A} \left( \frac{\mathbb{1} + S_s}{2} \right)$$




- Entanglement entropy: If the density matrix is proportional to a projector $\rho_A = C\mathcal{P}$

$$S_A^{(n)} = \frac{1}{1-n} \ln \mathrm{Tr} \rho_A^n = \frac{1}{1-n} \ln \left( C^{n-1} \mathrm{Tr} \rho_A \right) = -\ln C$$

# Frame Title

- For a multi-connected region with genus number $g$, there is a set of *web patch* configurations $\left\{\mathfrak{D}^{(1)}_{\mathrm{WP}}, \mathfrak{D}^{(2)}_{\mathrm{WP}}, \cdots, \mathfrak{D}^{(g)}_{\mathrm{WP}}\right\}$ such that $\mathrm{Tr}_A\left(\Sigma_{\mathfrak{D}\ominus\mathfrak{D}^{(n)}_{\mathrm{WP}}}\right) \neq 0$ if $\mathfrak{D} \subset A$
- In general, however, $\Sigma_{\mathfrak{D}\ominus\mathfrak{D}^{(n)}_{\mathrm{WP}}} \neq \Sigma_{\mathfrak{D}}\Sigma_{\mathfrak{D}^{(n)}_{\mathrm{WP}}}$, thus the reduced density matrix cannot be organized as before.
- The web patch is invalid in the fermion representation: $\mathrm{Tr}_{A,F}\left(\mathcal{M}_{\mathfrak{D}\ominus\mathfrak{D}^{(n)}_{\mathrm{WP}}}\right) = 0$